

BMAT202L - Probability and Statistics

Module 4 - Probability Distributions

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Introduction

- In this Module, we discuss Discrete and Continuous Distributions.
- They are as follows
 - 1 Binomial distribution;
 - 2 Poisson distributions;
 - 3 Normal distribution;
 - 4 Gamma distribution;
 - 5 Exponential distribution;
 - 6 Weibull distribution.

Bernoulli trial

- The experiment consists of ‘n’ repeated trials.
- Each trial results in an outcome that may be classified into 2 mutually exclusive categories such as a success or a failure.
- The probability of success is denoted by ‘p’ always remains constant from trial to trial.
- The repeated trials are independent.

Consider a trial which can be repeated for n times. In each trial, there are only 2 possible outcomes say a success or a failure. Let us denote the probability of success by p then the probability of failure q is $1 - p$. The probability of x success in n trials is $nC_x p^x q^{n-x}$. The probability distribution of the no. of success so obtained is called binomial distribution.

Binomial Distribution

A random variable X is said to follow binomial distribution if it assumes only non-negative values and its probability distribution is given by

$$P(X = x) = \begin{cases} nC_x p^x q^{n-x}, & x = 0, 1, 2 \dots n \\ 0, & \text{otherwise} \end{cases}$$

Conditions for Binomial Distribution

We get the Binomial Distribution under the following experimental conditions:

- ① The number of trials ' n ' is finite
- ② The trials are independent of each other
- ③ The probability of success ' p ' is same for each trial
- ④ Each trial must result in a success or a failure.

Characteristics of Binomial Distribution

i. Binomial distribution is a discrete distribution i.e., X can take values 0, 1, 2, ... n where ' n ' is finite .

ii. Constants of the distributions are:

Mean = np ; Variance = npq ; Standard deviation = $\sqrt{npq}$

$$\text{Skewness} = \frac{q-p}{\sqrt{npq}} ; \quad \text{Kurtosis} = \frac{1-6pq}{npq}$$

iii. It may have one or two modes.

iv. If $X \sim B(n_1, p)$ and $Y \sim B(n_2, p)$ and that X and Y are independent then $X+Y \sim B(n_1+n_2, p)$

v. If ' n ' independent trials are repeated N times the expected frequency of ' x ' successes is $N \times nC_x p^x q^{n-x}$

vi. If $p = 0.5$, the distribution is symmetric.

Binomial Distribution - MGF

The moment generating function of a binomial distribution

$$\begin{aligned}M_X(t) &= E[e^{tX}] = \sum_{x=0}^n e^{tx} p(x) \\&= \sum_{x=0}^n e^{tx} [n_{C_x} p^x q^{n-x}] \\&= n_{C_0} q^n + n_{C_1} e^t p q^{n-1} + n_{C_2} e^{2t} p^2 q^{n-2} + \dots + n_{C_n} e^{nt} p^n \\&= n_{C_0} q^n + n_{C_1} (e^t p) q^{n-1} + n_{C_2} (e^t p)^2 q^{n-2} + \dots + n_{C_n} (e^t p)^n \\&= (q + p e^t)^n\end{aligned}$$

Binomial Distribution - Mean

Find the mean and variance for a binomial distribution

$$\text{Mean} = \mu'_1$$

$$= \frac{d}{dt} [M_X(t)]_{t=0}$$

$$= \frac{d}{dt} [q + pe^t]^n_{t=0}$$

$$= [n(q + pe^t)^{n-1} p e^t]_{t=0}$$

$$= n(q + p)^{n-1} p$$

$$\text{Mean} = np \quad \because q + p = 1$$

Binomial Distribution - Variance

$$\begin{aligned}\mu'_2 &= \frac{d^2}{dt} [M_X(t)]_{t=0} \\&= [n(n-1)(q+pe^t)^{n-2} p^2 e^{2t} + n(q+pe^t)^{n-1} pe^t]_{t=0} \\&= n(n-1)(q+p)^{n-2} p^2 + np(q+p)^{n-1} \\&= n(n-1)p^2 + np \\ \mu'_2 &= n^2 p^2 - np^2 + np \\ \text{Variance} &= \mu'_2 - (\mu'_1)^2 \\&= n^2 p^2 - np^2 + np - n^2 p^2 \\ \text{Variance} &= npq\end{aligned}$$

Binomial Distribution - Cumulative

Let us suppose that the n trials of an experiment is repeated N times then distribution function of a binomial distribution is $N(nC_x p^x q^{n-x})$.

Cumulative distribution function of a binomial distribution

$$F_X(x) = \sum_{t \leq x} p(t) = \sum_{t=0}^x \binom{n}{t} p^t q^{n-t}$$

Properties of Binomial Distribution

- Binomial distribution is symmetrical if $p = q = 0.5$.
- It is skew symmetric if $p \neq q$.
- It is positively skewed if $p < 0.5$ and it is negatively skewed if $p > 0.5$.
- For Binomial distribution, variance is less than mean.

Problem 1

A fair coin is tossed 6 times. Find the probability that exactly 2 heads occurs.

Let X be a random variable follows binomial distribution with probability value $p = 1/2$ and $q = 1/2$

Prob: that exactcy 2 heads occur are as follows

$$\begin{aligned} P(X = 2) &= \binom{6}{x} p^x q^{n-x} \\ &= \binom{6}{2} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{6-2} \\ &= \frac{15}{64} \end{aligned}$$

Problem 2

Verify the following statement: The mean of a Binomial distribution is 12 and its standard deviation is 4.

$$\text{Mean: } np = 12$$

$$\text{SD} = \sqrt{npq} = 4$$

$$npq = 4^2 = 16, \quad \frac{np}{npq} = \frac{12}{16} = \frac{3}{4} \quad \text{which implies}$$
$$q = \frac{4}{3} > 1$$

Since $p + q$ cannot be greater than unity, the Statement is wrong

Problem 3

The probability that a student get the degree is 0.4 Determine the probability that out of 5 students (i) one will graduate (ii) at least one will graduate

Probability of getting a degree $p = 0.4$

$$\begin{aligned}\therefore q &= 1 - p \\ &= 1 - 0.4 \\ &= 0.6\end{aligned}$$

$$\begin{aligned}\text{(i) } P(\text{one will be a graduate}) &= P(X = 1) = {}^5C_1 (0.4)(0.6)^4 \\ &= 0.2592\end{aligned}$$

$$\begin{aligned}\text{(ii) } P(\text{at least one will be a graduate}) &= 1 - P(\text{none will be a graduate}) \\ &= 1 - {}^5C_0 (0.4)^0 (0.6)^5 \\ &= 1 - 0.0777 \\ &= 0.9222\end{aligned}$$

Problem 4

The mean of Binomials distribution is 20 and standard deviation is 4.
Find the parameters of the distribution.

The parameters of Binomial distribution are n and p

For Binomial distribution Mean = $np = 20$

Standard deviation = $\sqrt{npq} = 4$

$$\therefore npq = 16$$

$$\Rightarrow npq/np = 16/20 = 4/5$$

$$q = 4/5$$

$$\Rightarrow p = 1 - q = 1 - (4/5) = 1/5$$

Since $np = 20$

$$n = 20/p$$

$$n = 100$$

Problem 5

If on average, rain falls on 9 days in every thirty days, find the probability that rain will fall on at least two days of a given week.

Probability of raining on a particular day is given by $p = 9/30 = 3/10$ and

$$q = 1 - p = 7/10.$$

The BD is $P(X = x) = {}^nC_x p^x q^{n-x}$

There are 7 days in a week $P(X = x) = \binom{7}{x} \left(\frac{3}{10}\right)^x \left(\frac{7}{10}\right)^{7-x}$

The probability of raining for atleast 2 days is given by

$$\begin{aligned} P(X \geq 2) &= 1 - P(X < 2) \\ &= 1 - [P(X = 0) + P(X = 1)] \end{aligned}$$

Problem 5 Contd.

$$\begin{aligned}\text{Here, } P(X=0) &= \binom{7}{0} \left(\frac{3}{10}\right)^0 \left(\frac{7}{10}\right)^{7-0} \\ &= 0.0823\end{aligned}$$

$$\begin{aligned}\text{and } P(X=1) &= \binom{7}{1} \left(\frac{3}{10}\right) \left(\frac{7}{10}\right)^{7-1} \\ &= 0.2471\end{aligned}$$

Therefore the required probability = $1 - [P(x=0) + P(x=1)]$

$$= 1 - \{0.082 + 0.247\}$$

$$= 0.6706$$

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Problem 6

The sum and product of the mean and variance of a binomial distribution are 24 and 128. Find the distribution.

For Binomial Distribution the mean is np and variance is npq

Given values are $np + npq = 24$

$$np(1 + q) = 24 \quad (1)$$

Other term $np \times npq = 128$

$$n^2 p^2 q = 128 \quad (2)$$

From (1) we get $np = 24/(1+q)$ which implies $n^2 p^2 = (24/(1+q))^2$

Problem 6 Contd.

Substitute this value in equation (2) we get

$$\left(\frac{24}{1+q}\right)^2 q = 128 \quad \text{which implies } 9q = 2(1+2q+q^2)$$
$$(2q-1)(q-2) = 0$$

$$\text{Where } q = \frac{1}{2} \text{ and } p = \frac{1}{2}$$

Substitute in (1) we get $n = 32$

$$\text{Hence the binomial distribution } {}^{32}C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{32-x}$$

Problem 7

Ten coins are tossed simultaneously. Find the probability of getting (i) at least seven heads (ii) exactly seven heads (iii) at most seven heads.

X denote the number of heads appear

$$P(X = x) = {}^nC_x p^x q^{n-x}, x = 0, 1, 2, \dots, n$$

$$\text{Given: } p = P(\text{head}) = \frac{1}{2} \quad q = 1 - p = 1 - \frac{1}{2} = \frac{1}{2} \text{ and } n = 10$$

$$\therefore X \sim B\left(10, \frac{1}{2}\right)$$

$$= {}^{10}C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{10-x}$$

$$P(X = x) = \frac{{}^{10}C_x}{1024}$$

Problem 7 Contd.

(i) P (atleast seven heads)

$$\begin{aligned}P(X \geq 7) &= P(x = 7) + P(x = 8) + P(x = 9) + P(x = 10) \\&= \frac{10C_7}{1024} + \frac{10C_8}{1024} + \frac{10C_9}{1024} + \frac{10C_{10}}{1024} \\&= \frac{120}{1024} + \frac{45}{1024} + \frac{10}{1024} + \frac{1}{1024} \\&= \frac{176}{1024}\end{aligned}$$

(ii) P (exactly 7 heads)

$$P(x=7) = \frac{10C_7}{1024} = \frac{120}{1024}$$

Problem 7 Contd.

(iii) $P(\text{atmost 7 heads})$

$$= P(x \leq 7) = 1 - P(x > 7)$$

$$= 1 - \{P(x = 8) + P(x = 9) + P(x = 10)\}$$

$$= 1 - \left\{ \frac{10C_8}{1024} + \frac{10C_9}{1024} + \frac{10C_{10}}{1024} \right\}$$

$$= 1 - \frac{56}{1024}$$

$$= \frac{968}{1024}$$

Problem 8

With usual notation find p for Binomial random variable X if $n = 6$ and $9P(x = 4) = P(x = 2)$.

$$P(X = x) = {}^nC_x p^x q^{n-x}, x = 0, 1, 2, \dots, n$$

$$X \sim B(6, p) \Rightarrow P(X = x) = {}^6C_x p^x q^{6-x}$$

$$\text{Also } 9 \times P(X = 4) = P(X = 2)$$

$$\Rightarrow 9 \times {}^6C_4 p^4 q^2 = {}^6C_2 p^2 q^4$$

$$\Rightarrow 9 p^2 = q^2$$

$$\Rightarrow 3p = q \quad \text{as } p, q > 0$$

$$3p = 1 - p$$

$$4p = 1$$

$$\Rightarrow p = \frac{1}{4} = 0.25$$

Problem 9

From a pack of 52 cards 4 cards are drawn one after another with replacement. Find the mean and variance of the distribution of the number of kings.

Success X =event of getting king in a draw

p =probability of getting king in a single trial

$$\begin{aligned} p &= \frac{4}{52} \\ &= \frac{1}{13} \end{aligned}$$

This is constant for each trial.

Hence, it is a binomial distribution with $n=4$ and $p = \frac{1}{13}$

$$\begin{aligned} \text{Mean} &= np &= 4 \times \frac{1}{13} &= \frac{4}{13} \\ \text{Variance} &= npq &= \frac{4}{13} \times \frac{12}{13} &= \frac{48}{169} \end{aligned}$$

Problem 10

In a street of 200 families, 40 families purchase the Hindu newspaper. Among the families a sample of 10 families is selected, find the probability that

- i. Only one family purchase the news paper
- ii. No family purchases the newspaper
- iii. Not more than one family purchase it

$$X \sim B(n, p)$$

$$P(X = x) = {}^nC_x p^x q^{n-x}, x = 0, 1, 2, \dots, n$$

Let X denote the number of families purchasing Hindu Paper

p = Probability of their family purchasing the Hindu

$$p = \frac{40}{200} = \frac{1}{5}$$

$$q = \frac{4}{5}$$

$$n = 10$$

Problem 10 Contd.

(i) Only one family purchase the Hindu

$$\begin{aligned}P(X=1) &= {}^{nC_1} p^1 q^{n-1} \\&= {}^{10C_1} \times \frac{1}{5} \times \left(\frac{4}{5}\right)^9 \\&= 2 \times \left(\frac{4}{5}\right)^9\end{aligned}$$

(ii) No family purchasing the Hindu

$$\begin{aligned}P(X=0) &= {}^{10C_0} \left(\frac{1}{5}\right)^0 \left(\frac{4}{5}\right)^{10} \\&= \left(\frac{4}{5}\right)^{10}\end{aligned}$$

Problem 10 Contd.

(iii) Not more than one family purchasing The Hindu means that $X \leq 1$

$$\begin{aligned}P(X \leq 1) &= P[x=0] + P[x=1] \\&= {}^{10}C_0 \left(\frac{1}{5}\right)^0 \left(\frac{4}{5}\right)^{10} + {}^{10}C_1 \left(\frac{1}{5}\right)^1 \left(\frac{4}{5}\right)^9 \\&= \left(\frac{4}{5}\right)^{10} + 10 \times \left(\frac{1}{5}\right)^1 \left(\frac{4}{5}\right)^9 \\&= \left(\frac{4}{5}\right)^9 \left[\left(\frac{4}{5}\right) + 2 \right] \\&= \left(\frac{4}{5}\right)^9 \left(\frac{14}{5}\right)\end{aligned}$$

Problem 11

10 coins are thrown simultaneously. Find the probability of getting at least 7 heads.

Trial is tossing a coin

Success event is getting head

Take p = probability of getting head = $\frac{1}{2}$

q = probability of getting tail = $\frac{1}{2}$

$n = 10$

In a binomial distribution probability of getting x heads in a throw of 10 coins is $P(X = x)$

$$P(X = x) = n_c_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{10-x}$$

Problem 11 Contd.

Probability of getting at least 7 heads = $P(X = 7) + P(X = 8) + P(X = 9) + P(X = 10)$

$$\begin{aligned} &= {}^{10}C_7 \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^3 + {}^{10}C_8 \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^2 + {}^{10}C_9 \left(\frac{1}{2}\right)^9 \left(\frac{1}{2}\right)^1 + {}^{10}C_{10} \left(\frac{1}{2}\right)^{10} \\ &= \left(\frac{1}{2}\right)^{10} [120 + 45 + 10 + 1] \\ &= \left(\frac{1}{2}\right)^{10} [176] \\ &= \frac{11}{64} \end{aligned}$$

Problem 12

Out of 800 families with 4 children each, how many families would be expected to have

(i) 2 boys & 2 girls

(ii) At least one boy

(iii) At most 2 girls

Let us consider birth of each child as a trial

In the given problem, $n = 4, N = 800$

Let us assume birth of a boy as a success event

probability of success $p = \frac{1}{2}$ then $q = \frac{1}{2}$

The distribution function for the given problem is

No. of families having child = $N[n_{C_x} p^x q^{n-x}]$

$$= 800 \left[{}^4C_x \left(\frac{1}{2} \right)^x \left(\frac{1}{2} \right)^{4-x} \right]$$

Problem 12 Contd.

(i) No. of families having 2 boys and 2 boys = $800[P[X = 2]]$

$$= 800 \left[{}^4C_2 \left(\frac{1}{2} \right)^2 \left(\frac{1}{2} \right)^2 \right]$$

$$= 800 \times 6 \times \frac{1}{4} \times \frac{1}{4}$$

$$= 300$$

(ii) No. of families having atleast one boy = $800[P[X = 1] + P[X = 2] + P[X = 3] + P[X = 4]]$

$$= 800[1 - P[X = 0]]$$

$$= 800 \left[1 - {}^4C_0 \left(\frac{1}{2} \right)^0 \left(\frac{1}{2} \right)^4 \right]$$

$$= 750$$

(iii) No. of families having atmost 2 girls = $800[P(\text{no girl}) + P(1 \text{ girl}) + P(2 \text{ girls})]$

$$= 800[P[X = 4] + P[X = 3] + P[X = 2]]$$

$$= 800[1 - P[X = 0]]$$

$$= 800 \left[{}^4C_2 \left(\frac{1}{2} \right)^4 \left(\frac{1}{2} \right)^0 + {}^4C_3 \left(\frac{1}{2} \right)^3 \left(\frac{1}{2} \right) + {}^4C_2 \left(\frac{1}{2} \right)^2 \left(\frac{1}{2} \right)^2 \right]$$

$$= 800 \times \frac{1}{16} [1 + 4 + 6]$$

$$= 550$$

Problem 13

The probability that a patient recovers from a disease is 0.3. If 18 people are affected from this what is the probability that

(i) At least 10 people survive

(ii) Exactly 6 survive

(iii) 4 to 7 people

The event success is a patient who recovers from a disease

$$p = 0.3, q = 0.7, n = 18$$

By applying binomial probability law, the probability of x people (survive) recover from disease is

$$P(X = x) = {}^{18}C_x (0.3)^x (0.7)^{18-x}$$

Problem 13 Contd.

(i) P(at least 10 people survive)

$$\begin{aligned}P(X \geq 10) &= {}^{18}C_{10} (0.3)^{10}(0.7)^8 + {}^{18}C_{11} (0.3)^{11}(0.7)^7 + {}^{18}C_{12} (0.3)^{12}(0.7)^6 + {}^{18}C_{13} (0.3)^{13}(0.7)^5 \\&\quad + {}^{18}C_{14} (0.3)^{14}(0.7)^4 + {}^{18}C_{15} (0.3)^{15}(0.7)^3 + {}^{18}C_{16} (0.3)^{16}(0.7)^2 + {}^{18}C_{17} (0.3)^{17}(0.7)^1 \\&\quad + {}^{18}C_{18} (0.3)^{18} \\&= 0.021\end{aligned}$$

(ii) P(exactly 6 people survive)

$$\begin{aligned}P(X = 6) &= {}^{18}C_6 (0.3)^6(0.7)^{12} \\&= 0.1873\end{aligned}$$

Problem 13 Contd.

(iii) P(4 to 7 people)

$$P(4 \leq X \leq 7) = P(X = 4) + P(X = 5) + P(X = 6) + P(X = 7)$$

$$= 18c_4 (0.3)^4(0.7)^{14} + 18c_5 (0.3)^5(0.7)^{13} + 18c_6 (0.3)^6(0.7)^{12} + 18c_7 (0.3)^7(0.7)^{11}$$

$$= 0.168 + 0.201 + 0.187 + 0.1376$$

$$= 0.6947$$

Problem 14

Find the mean and standard deviation whose moment generating function is $(0.4e^t + 0.6)$

$$M_X(t) = (0.4e^t + 0.6)$$

W.K.T $M_X(t)$ for binomial distribution = $(q + pe^t)^n$

$$\Rightarrow n = 1, q = 0.6, p = 0.4$$

$$\text{Mean} = np = 0.4$$

$$\text{Standard deviation} = \sqrt{npq} = \sqrt{0.24}$$

Problem 15

If X follows $B\left(25, \frac{1}{5}\right)$ find $P(X < \mu - 2\sigma)$

$$n = 25, \quad p = \frac{1}{5} \Rightarrow q = \frac{4}{5}$$

$$\mu = np = 25 \times \frac{1}{5} = 5$$

$$\text{Var} = \sigma^2 = npq = 25 \times \frac{1}{5} \times \frac{4}{5} = 4$$

$$\sigma = 2$$

$$P(X < 5 - 4) = P(X < 1) = P(X = 0) = {}^{n}C_0 p^0 (q)^{n-0}$$

$$= {}^{25}C_0 \left(\frac{1}{5}\right)^0 \left(\frac{4}{5}\right)^{25}$$

$$= \left(\frac{4}{5}\right)^{25}$$

Fitting of Binomial Distribution

When a Binomial distribution is to be fitted to an observed data the following procedure is adopted:-

- (i) Find mean $\bar{x} = \frac{\sum fx}{\sum f} = np$
- (ii) Find $p = \frac{\bar{x}}{n}$
- (iii) Find $q = 1 - p$
- (iv) Write the probability mass function : $P(x) = nC_x p^x q^{n-x} \quad x = 0, 1, 2, \dots, n$
- (v) Put $x = 0$; find $P(0) = nC_0 p^0 q^{n-0} = q^n$
- (vi) Find the expected frequency of $X = 0$ i.e., $F(0) = N \times P(0)$, where $N = \sum f_i$
- (vii) The other expected frequencies are obtained by using the recurrence formula
$$F(x+1) = \frac{n-x}{x+1} \times \frac{p}{q} \times F(x)$$

Problem 16

A set of three similar coins are tossed 100 times with the following results

Number of heads	0	1	2	3
Frequency	36	40	22	2

Fit a binomial distribution and estimate the expected frequencies.

Problem 16 Contd.

x	f	fx
0	36	0
1	40	40
2	22	44
3	2	44
Total	100	90

Problem 16 Contd.

$$(i) \text{ Mean } \bar{x} = \frac{\sum fx}{\sum f} = \frac{90}{100} = 0.9$$

$$(ii) p = \frac{\bar{x}}{n} = \frac{0.9}{3} = 0.3$$

$$(iii) q = 1 - p = 1 - 0.3 = 0.7$$

$$(iv) P(x) = {}^nC_x p^x q^{n-x} = {}^3C_x 0.3^x 0.7^{3-x}$$

$$(v) P(0) = {}^3C_0 0.3^0 (0.7)^{3-0} = 0.7^3 = 0.343$$

Problem 16 Contd.

$$(vi) \quad F(0) = N \times P(0) = 100 \times 0.343 = 34.3$$

$$(vii) \quad F(x+1) = \frac{n-x}{x+1} \times \frac{p}{q} \times F(x)$$

$$\therefore \quad F(1) = F(0+1) = \frac{3-0}{0+1} \times \frac{0.3}{0.7} \times 34.3 = 44.247$$

$$F(2) = F(1+1) = \frac{3-1}{1+1} \times \frac{0.3}{0.7} \times 44.247 = 19.03$$

$$F(3) = F(2+1) = \frac{3-2}{2+1} \times \frac{0.3}{0.7} \times 19.03 = 2.727$$

Problem 16 Contd.

(i) The fitted binomial distribution is

$$P(X = x) = {}^3C_x 0.3^x 0.7^{(3-x)}, \quad x = 0, 1, 2, 3$$

(ii) The expected frequencies are :

x	0	1	2	3	Total
Observed frequencies (O_i)	36	40	22	2	100
Expected Frequencies (E_i)	34	44	19	3	100

Practice Problems I

- ① The incidence of an occupational disease in an industry is such that the workers have a 20% chance of suffering from it. What is the probability that out of 6 workers chosen at random, four more will suffer from the disease? (**Ans: 0.0175**)
- ② If the probability of a defective bolt is $\frac{1}{8}$, find the mean and variance of the distribution of defective bolts of 640. (**Ans: 80, 70**)
- ③ In 256 sets of 12 tosses of a coin, in how many cases one can expect 8 heads and 4 tails. (**Ans: 31**)
- ④ If the probability of success is $\frac{1}{20}$, how many trials are necessary in order that the probability of at least one success is just greater than $\frac{1}{2}$. (**Ans: 14**)
- ⑤ Six dice are thrown 729 times. How many times do you expect at least three dice to show a 5 or 6? (**Ans: 233**)

Practice Problems II

- ⑥ The mean and variance of a binomial distribution are 4 and $\frac{4}{3}$ respectively. Find $P(X \geq 1)$. (**Ans: 0.9986**)
- ⑦ The mean and variance of a binomial variable X with parameters n and p are 16 and 8. Find $P(X \geq 1)$ and $P(X > 2)$. (**Ans: $1 - \frac{1}{2^{32}}, 0.9999$**)
- ⑧ In a binomial distribution consisting of n independent trials, probabilities of 1 and 2 are 0.4096 and 0.2048 respectively. Find the parameter p of the distribution. (**Ans: 0.2**)
- ⑨ The probability of a man hitting a target is $\frac{1}{3}$.
 - ① If he fires 5 times, what is the probability of him hitting the target at least twice? (**Ans: 0.5391**)
 - ② How many times must he fire so that the probability of him hitting the target at least once is more than 90%? (**Ans: $n = 6$**)

Practice Problems III

- 10 The difference between the mean and variance of a binomial distribution is 1 and the difference between their squares is 11. Find n . (**Ans: 36**)

- 11 Fit a binomial distribution to the following data:

x	0	1	2	3	4	5	6
f	13	25	52	58	32	16	4

- 12 Fit a binomial distribution to the following data:

x	0	1	2	3	4	5
f	2	14	20	34	22	8

- 13 Four coins are tossed 160 times. The number of times x heads occur is given below:

x	0	1	2	3	4
f	8	34	69	43	6

Fit a binomial distribution to this data.

Poisson Distribution

The poisson distribution occurs often as module which do not have the natural upper bound. This distribution occurs when there are events which do not occur as outcomes of a definite number of trials.

Generally, if $n > 30$ and $p \leq 0.05$ we apply poisson distribution as an approximation to binomial.

Definition:

The probability distribution function of a poisson random variable is given by

$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$ where $x = 0, 1, 2, 3 \dots \infty$, λ is the parameter and $\lambda > 0$.

Poisson Distribution - M. G. F.

Find the moment generating function, mean and variance for poisson distribution.

$$\begin{aligned}M_X(t) &= E[e^{tX}] = \sum_{x=0}^{\infty} e^{tx} \cdot \frac{e^{-\lambda} \lambda^x}{x!} \\&= e^{-\lambda} \sum_{x=0}^{\infty} \frac{e^{-tx} \lambda^x}{x!} \\&= e^{-\lambda} \left[1 + \frac{e^t \lambda}{1!} + \frac{e^{2t} \lambda^2}{2!} + \frac{e^{3t} \lambda^3}{3!} + \dots \right] \\&= e^{-\lambda} \left[1 + \frac{e^t \lambda}{1!} + \frac{(e^t \lambda)^2}{2!} + \frac{(e^t \lambda)^3}{3!} + \dots \right] \\&= e^{-\lambda} \left[e^{\lambda e^t} \right]\end{aligned}$$

$$M_X(t) = e^{\lambda(e^t - 1)}$$

Poisson Distribution - Mean and Variance

$$\text{Mean} = \frac{d}{dt} [M_X(t)]_{t=0}$$

$$= \left[e^{\lambda(e^t-1)} \lambda e^t \right]_{t=0}$$

$$= \lambda$$

$$\text{Variance} = \mu'_2 - (\mu'_1)^2$$

$$\mu'_2 = \frac{d^2}{dt^2} [M_X(t)]_{t=0} = \lambda \left[e^t e^{\lambda(e^t-1)} + e^{\lambda(e^t-1)} \lambda e^t e^t \right]_{t=0} = \lambda[1 + \lambda]$$

$$\text{Variance} = \lambda$$

Characteristics of Poisson Distribution

- Poisson distribution is a discrete distribution *i.e.*, X can take values $0, 1, 2, \dots$
- p is small, q is large and n is indefinitely large.
- Mean = λ , Variance = λ , Standard deviation = $\sqrt{\lambda}$, Skewness = $\frac{1}{\sqrt{\lambda}}$,
Kurtosis = $\frac{1}{\lambda}$
- It may have one or two modes
- If X and Y are two independent Poisson variates, $X + Y$ is also a Poisson variate.
- Poisson distribution is positively skewed.
- It is leptokurtic.

Limiting Concept

Prove that the poisson distribution is a limiting case of binomial distribution.

W.K.T. the distribution function for binomial random variable is

$$\begin{aligned}P(X = x) &= n C_x p^x q^{n-x} \\&= \frac{n!}{(n-x)! x!} p^x q^{n-x} \\&= \frac{n!}{(n-x)! x!} p^x (1-p)^{n-x}\end{aligned}$$

$$\text{Take } np = \lambda \Rightarrow p = \frac{\lambda}{n}$$

$$P(X = x) = \frac{[n(n-1)\dots(n-x+1)(n-x)\dots 2.1]}{[(n-x)(n-x-1)\dots 2.1] x!} \left(\frac{\lambda}{n}\right)^x \left[1 - \frac{\lambda}{n}\right]^{n-x}$$

Limiting Concept Contd.

$$\begin{aligned} &= \frac{[n(n-1)\dots(n-[x-1])]}{x!} \left(\frac{\lambda}{n}\right)^x \left[1 - \frac{\lambda}{n}\right]^{n-x} \\ &= \frac{[n(n-1)\dots(n-[x-1])]}{x!} \frac{\lambda^x}{n^x} \left[1 - \frac{\lambda}{n}\right]^n \left[1 - \frac{\lambda}{n}\right]^{-x} \\ &= \left(\frac{n}{n}\right) \left(1 - \frac{1}{n}\right) \dots \left(1 - \frac{[x-1]}{n}\right) \frac{\lambda^x}{x!} \left[1 - \frac{\lambda}{n}\right]^n \left[1 - \frac{\lambda}{n}\right]^{-x} \end{aligned}$$

Tending $n \rightarrow \infty$

$$= 1 \frac{\lambda^x}{x!} \lim_{n \rightarrow \infty} \left[1 - \frac{\lambda}{n}\right]^n \lim_{n \rightarrow \infty} \left[1 - \frac{\lambda}{n}\right]^{-x}$$

$$\lim_{n \rightarrow \infty} \left[1 - \frac{\lambda}{n}\right]^n = e^{-\lambda}$$

If power is not n the value for particular limit is

$$P(X = x) = \frac{\lambda^x}{x!} e^{-\lambda}$$

Problem 1

In a Poisson distribution the first probability term is 0.2725. Find the next Probability term.

$$\text{Given that } p(0) = 0.2725$$

$$\frac{e^{-\lambda} \lambda^0}{0!} = 0.2725$$

$$\Rightarrow e^{-\lambda} = 0.2725 \quad (\text{by using exponent table})$$

$$\lambda = 1.3$$

$$\therefore p(X=1) = e^{-1.3} (1.3) / 1!$$

$$= e^{-1.3} (1.3)$$

$$= 0.2725 \times 1.3$$

$$= 0.3543$$

Problem 2

An insurance company has discovered that only about 0.1 percent of the population is involved in a certain type of accident each year. If its 10,000 policy holders were randomly selected from the population, what is the probability that not more than 5 of its clients are involved in such an accident next year? ($e^{-10} = 0.000045$)

Problem 2

p = probability that a person will involve in an accident in a year

$$= 0.1/100 = 1/1000$$

given $n = 10,000$

$$\text{so, } \lambda = np = 10000 (1/1000) = 10$$

Probability that not more than 5 will involve in such an accident in a year

$$P(X < 5) = P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5)$$

$$= e^{-10} \left[1 + \frac{10}{1!} + \frac{10^2}{2!} + \frac{10^3}{3!} + \frac{10^4}{4!} + \frac{10^5}{5!} \right]$$

$$= 0.06651$$

Problem 3

In a book of 520 pages, 390 typo-graphical errors occur. Assuming Poisson law for the number of errors per page, find the probability that a random sample of 5 pages will contain no error.

The average number of typographical errors per page in the book is given by $\lambda = \frac{390}{520} = 0.75$.

Hence using Poisson probability law, the probability of x errors per page is given by

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-0.75} (0.75)^x}{x!}, x = 0, 1, 2, \dots$$

The required probability that a random sample of 5 pages will contain no error is given by

$$P(X = 5) = \frac{e^{-0.75} (0.75)^5}{5!} = e^{-3.75}$$

Problem 4

When counting red blood cells, a square grid is used, over which a drop of blood is evenly distributed. Under the microscope an average of 8 erythrocytes are observed per single square. What is the probability that exactly 5 erythrocytes are found in one square?

Let X be a random variable follows poisson distribution with number of erythrocytes.

Hence, Mean $\lambda = 8$ erythrocytes/single square

$P(\text{exactly 5 erythrocytes are in one square}) = P(X = 5) =$

$$\begin{aligned} &= \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-8} 8^5}{5!} \\ &= \frac{0.000335 \times 32768}{120} \\ &= 0.0916 \end{aligned}$$

The probability that exactly 5 erythrocytes are found in one square is 0.0916. i.e there are 9.16% that exactly 5 erythrocytes are found in one square.

Problem 5

Assuming one in 80 births is a case of twins, calculate the probability of 2 or more sets of twins on a day when 30 births occur.

Let x denotes the set of twins on a day

$$P(\text{twin birth}) = p = 1/80 = 0.0125 \text{ and } n = 30$$

$$\text{The value of mean } \lambda = np = 30 \times 0.0125 = 0.375$$

$$\text{Hence, } X \text{ follows poisson distribution with } p(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

Problem 5 Contd.

The probability is

$$P(2 \text{ or more}) = 1 - [p(x=0) + p(x=1)] =$$

$$= 1 - \left[\frac{e^{-0.375} (0.375)^0}{0!} + \frac{e^{-0.375} (0.375)^1}{1!} \right]$$

$$= 1 - e^{-0.375} [1 + 0.375]$$

$$= 1 - (0.6873 \times 1.375)$$

$$= 0.055$$

Problem 6

If 2% of electric bulbs manufactured by a certain company are defective find the probability that in a sample of 200 bulbs (i) less than 2 bulbs are defective (ii) more than 3 bulbs are defective. [$e^{-4} = 0.0183$].

Let X denote the number of defective bulbs

$$X \sim P(\lambda)$$

$$\therefore P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad x = 0, 1, 2, \dots, \infty$$

$$\text{Given } p = P(\text{a defective bulb}) = 2\% = \frac{2}{100} = 0.02$$

$$n = 200$$

$$\therefore \lambda = np = 200 \times 0.02 = 4$$

$$\therefore P(X = x) = \frac{e^{-4} 4^x}{x!}, \quad x = 0, 1, 2, \dots, \infty$$

Problem 6 Contd.

(i) P (less than 2 bulbs are defective)

$$= P(X < 2)$$

$$= P(x = 0) + P(x = 1)$$

$$= \frac{e^{-4} \cdot 4^0}{0!} + \frac{e^{-4} \cdot 4^1}{1!}$$

$$= e^{-4}(1 + 4)$$

$$= 0.0183 \times 5$$

$$= 0.0915$$

Problem 6 Contd.

(ii) P (more than 3 defectives)

$$= P(X > 3)$$

$$= 1 - P(X \leq 3)$$

$$= 1 - \{ P(x = 0) + P(x = 1) + P(x = 2) + P(x = 3) \}$$

$$= 1 - \left\{ \frac{e^{-4} \cdot 4^0}{0!} + \frac{e^{-4} \cdot 4^1}{1!} + \frac{e^{-4} \cdot 4^2}{2!} + \frac{e^{-4} \cdot 4^3}{3!} \right\}$$

$$= 1 - e^{-4} \{1 + 4 + 8 + 10.667\}$$

$$= 1 - 0.0183 \times 23.667$$

$$= 0.567$$

Problem 7

In a Poisson distribution $3P(X = 2) = P(X = 4)$. Find its parameter λ .

The *pmf* of Poisson distribution is $P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$, $x = 0, 1, 2, \dots, \infty$,

Given $3P(X = 2) = P(X = 4)$

$$3 \cdot \frac{e^{-\lambda} \lambda^2}{2!} = \frac{e^{-\lambda} \cdot \lambda^4}{4!}$$

$$\lambda^2 = \frac{3 \times 4!}{2!} = 36$$

$$\therefore \lambda = 6 \text{ as } \lambda > 0$$

Problem 8

If there are 400 errors in a book of 1000 pages, find the probability that a randomly chosen page from the book has exactly 3 errors.

Let X denote the number of errors in pages

Let X denote the number of errors in pages

$$X \sim P(\lambda)$$

$$\therefore P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad x = 0, 1, 2, \dots, \infty$$

$$\text{The average number of errors per page} = \frac{400}{1000}$$

$$\text{i.e., } \lambda = \frac{400}{1000} = 0.4$$

$$\begin{aligned} P(X=3) &= \frac{e^{-\lambda} \lambda^3}{3!} = \frac{e^{-0.4} (0.4)^3}{3 \times 2 \times 1} \\ &= \frac{0.6703 \times 0.064}{6} \\ &= 0.00715 \end{aligned}$$

Problem 9

The probability of safety pin manufactured by a firm to be defective is 0.04. (i) Find the probability that a box containing 100 such pins has one defective pin . (ii) Among 200 such boxes, how many boxes will have no defective pin

Let X denote the number boxes with defective pins

$$X \sim P(\lambda)$$

$$\therefore P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad x = 0, 1, 2, \dots, \infty$$

$$p = 0.04$$

$$n = 100$$

$$\lambda = np = 4$$

Problem 9 Contd.

$$\begin{aligned} \text{(i)} \quad P(X=1) &= \frac{e^{-\lambda} \lambda}{1!} = e^{-4}(4) = 0.0183 \times 4 \\ &= 0.0732 \end{aligned}$$

$$\text{(ii)} \quad P(X=0) = \frac{e^{-\lambda} \lambda^0}{0!} = e^{-4} = 0.0183$$

$$\begin{aligned} \text{Number of boxes having no defective pin} &= 200 \times 0.0183 \\ &= 3.660 \\ &= 4 \end{aligned}$$

Problem 10

The number of accidents in a year to taxi drivers in a city follows poisson distribution with mean 3 out of 1000 taxi drivers, find approximate no. of drivers

(i) With no accidents in a year

(ii) More than 3 accidents in a year

Let the success event be no. of accidents of taxi drivers

$$\text{Mean} = \lambda = 3$$

$$P(X = 0) = \frac{e^{-\lambda} \lambda^0}{0!}$$

$$= \frac{e^{-3} 3^0}{0!} = 0.0498$$

Problem 10 contd.

Out of thousand drivers the no. of drivers with no accidents in a year is

$$1000 \times P(X = 0) = 1000 \times 0.0498$$

$$= 49.8$$

$$= 50 \text{ drivers}$$

$$\text{(ii) } P(X > 3) = 1 - P(X \leq 3)$$

$$= 1 - e^{-3} \left[\frac{3^0}{0!} + \frac{3}{1!} + \frac{3^2}{2!} + \frac{3^3}{3!} \right]$$

$$= 1 - e^{-3} [13]$$

$$= 1 - 0.6472$$

$$= 0.3527$$

Problem 10 contd.

No. of drivers with more than 3 accidents is

$$\begin{aligned} 1000 \times P(X > 3) &= 1000 \times 0.3527 \\ &= 353 \text{ drivers} \end{aligned}$$

Fitting of Poisson Distribution

When a Poisson distribution is to be fitted to an observed data the following procedure is adopted:

- (i) Find the mean: $\bar{x} = \frac{\sum fx}{\sum f}$
- (ii) Poisson parameter = $\lambda = \bar{x}$
- (iii) Probability mass function is: $P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad x = 0, 1, 2, \dots$
- (iv) Put $X = 0$ and find $P(0) = \frac{e^{-\lambda} \lambda^0}{0!} = e^{-\lambda}$
- (v) Expected frequency for $x = 0$ is $F(0) = N \times P(0)$, $\sum f_i = N$
- (vi) Other expected frequencies can be found using
$$F(x + 1) = \frac{\lambda}{x + 1} \times F(x) \text{ for } x = 0, 1, 2, \dots$$

Problem 11

The following mistakes per page were observed in a book

Number of Mistakes (per page)	0	1	2	3	4
Number of pages	211	90	19	5	0

Fit a Poisson distribution and estimate the expected frequencies.

Problem 11 Contd.

x	f	fx
0	211	0
1	90	90
2	19	38
3	5	15
4	0	0
Total	325	143

Problem 11 Contd.

$$(i) \text{ Mean } \bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{143}{325} = 0.44$$

$$(ii) \lambda = \bar{x} = 0.44$$

$$(iii) P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-0.44} (0.44)^x}{x!}$$

$$(iv) P(0) = \frac{e^{-0.44} \times 0.44^0}{0!} = e^{-0.44} = 0.6440 \text{ (from the Poisson table)}$$

$$(v) F(0) = N \times P(0) = 325 \times 0.6440 = 209.43$$

Problem 11 Contd.

$$(vi) \quad F(x+1) = \frac{\lambda}{x+1} F(x)$$

$$F(1) = F(0+1) = \frac{0.44}{0+1} \times 209.43 = 92.15$$

$$F(2) = F(1+1) = \frac{0.44}{1+1} \times 92.15 = 20.27$$

$$F(3) = F(2+1) = \frac{0.44}{2+1} \times 2.972$$

$$F(4) = F(3+1) = \frac{0.44}{3+1} \times 2.97 = 0.33$$

Problem 11 Contd.

- (1). Fitted Poisson distribution is $P(X = x) = \frac{e^{-0.44} 0.44^x}{x!}$, $x = 0, 1, 2, \dots$
- (2). Expected frequencies are given below :

x	0	1	2	3	4	Total
Observed frequencies (O_i)	211	90	19	5	0	325
Expected Frequencies (E_i)	210	92	20	3	0	325

Problem 12

The probability of getting no misprint in a page of a book is e^{-4} . What is the probability that a page contains more than 2 prints.

Success event is getting a misprint

$$P(X = 0) = e^{-4}$$

$$\frac{e^{-\lambda} \lambda^0}{0!} = e^{-4}$$

$$\lambda = 4$$

$$P(X > 2) = 1 - P(X \leq 2)$$

$$= 1 - e^{-4} \left[1 + \frac{4^1}{1!} + \frac{4^2}{2!} \right]$$

$$= 0.7619$$

Problem 13

Fit a poisson distribution to the following data

x	f
0	142
1	156
2	69
3	27
4	5
5	1

Problem 13 Contd.

x	f	fx	Poisson frequency
0	142	0	147
1	156	156	147
2	69	138	74
3	27	81	25
4	5	20	6
5	1	5	1
	400	400	400

The poisson frequency are obtained by using the

formula $\frac{N e^{-\lambda} \lambda^x}{x!}$ where $N = \sum f$ and $\lambda = \frac{\sum fx}{\sum f}$

$$N = 400, \lambda = \frac{400}{400} = 1$$

$$\frac{N e^{-\lambda} \lambda^x}{x!} = \frac{400 e^{-1} 1^x}{x!}$$

X varies from 0 to 5

$$x = 0 \Rightarrow 400 e^{-1} = 147$$

$$x = 1 \Rightarrow 400 e^{-1} = 147$$

$$x = 2 \Rightarrow \frac{400 e^{-2}}{2} = 74$$

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Problem 14 TRY!

Fit a poisson distribution to the following data

x	f
0	142
1	156
2	69
3	27
4	5
5	1

Practice Problems I

- ① If the probability that an individual suffers a bad reaction from a certain injection is 0.001, determine the probability that out of 2000 individuals i.) exactly 3 ii.) more than 2 individuals iii.) None iv.) more than one individual suffer a bad reaction. **Ans: i.) 0.1804, ii.) 0.333, iii.) 0.1353, iv.) 0.5940**
- ② A hospital switch board receives an average of 4 emergency calls in a 10 minute interval. What is the probability that i.) there are at most 2 emergency calls in a 10 minute interval, ii.) there are exactly 3 emergency calls in a 10 minute interval. **Ans: i.) 0.2381, ii.) 0.1954**
- ③ If a bank received on the average 6 bad cheques per day, find the probability that it will received 4 bad cheques on any given day. **Ans: 0.1339**

Practice Problems II

- ④ Using Poisson's distribution, find the probability that the ace of spades will be drawn from a pack of well shuffled cards at least once in 104 consecutive trials. **Ans: 0.8647**
- ⑤ Average number of accidents on any day on a national highway is 1.8. Determine the probability that the number of accidents are i.) at least one, ii.) at most one. **Ans: i.) 0.0.8347, ii.) 0.4628**
- ⑥ A distributor of bean seeds determines from extensive tests that 5% of large batch of seeds will not germinate. He sells the seeds in packets of 200 and guarantees 90% germination. Determine the probability that a particular packet will violate the guarantee. **Ans: 0.0016**
- ⑦ After correcting 50 pages, the proof reader finds that there are on the average of 2 errors per 5 pages. How many pages could one expect with 0 error, 1 error and at least 3 errors in 1000 pages of the first print of the book? **Ans: 8**

Practice Problems III

- 8 The distribution of typing mistakes committed by a typist is given below. Assuming the distribution to be Poisson, find the expected frequencies

x	0	1	2	3	4	5
$f(x)$	42	33	14	6	4	1

- 9 Fit a Poisson distribution for the following data and calculate the expected frequencies.

x	0	1	2	3	4
$f(x)$	109	65	22	3	1

Normal Distribution / Gaussian Distribution

A continuous random variable X is said to have normal distribution with the

parameters μ, σ^2 , if its density function is given by $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$, $-\infty < x <$

0 and $\sigma > 0$. The normal curve is bell shaped curve and it is symmetrical with respect to the axis $x = \mu$. The smaller values of σ gives the greater max value of $f(x)$ so the quantity σ controls the dispersion values of random variable X of $\mu > 0$, the curve is shifted to the right for $\mu < 0$, the curve shift to the left.

Normal Distribution - M. G. F.

$$\begin{aligned}M_X(t) &= E[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \\&= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx\end{aligned}$$

$$\text{Take } k = \frac{x-\mu}{\sigma} \quad dk = \frac{dx}{\sigma}$$

When $x = \infty, k = \infty$

$$x = -\infty, k = -\infty$$

$$M_X(t) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{t(\mu+k\sigma)} e^{-\frac{k^2}{2}} \sigma dk$$

Normal Distribution - M. G. F.

$$= \frac{1}{\sqrt{2\pi}} e^{\mu t} \int_{-\infty}^{\infty} e^{kt\sigma} e^{-\frac{k^2}{2}} dk$$

$$= \frac{e^{\mu t}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(\frac{k^2}{2} - kt\sigma\right)} dk$$

Note: $\int_{-\infty}^{\infty} e^{-\frac{\omega^2}{2}} d\omega = \sqrt{2\pi}$

$$M_X(t) = \frac{e^{\mu t}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\left(\frac{k^2}{2} - kt\sigma + \frac{t^2\sigma^2}{2}\right)} e^{\frac{t^2\sigma^2}{2}} dk$$

$$= \frac{e^{\mu t}}{\sqrt{2\pi}} e^{\frac{t^2\sigma^2}{2}} \int_{-\infty}^{\infty} e^{-\left(\frac{k}{\sqrt{2}} - \frac{t\sigma}{\sqrt{2}}\right)^2} dk$$

Assume $k - t\sigma = \omega$

$$dk = d\omega$$

When $k = -\infty$, $\omega = -\infty$

$$k = \infty, \quad \omega = \infty$$

$$a = \frac{k}{\sqrt{2}}$$

$$2ab = kt\sigma$$

$$2 \cdot \frac{k}{\sqrt{2}} b = kt\sigma$$

$$b = \frac{t\sigma}{\sqrt{2}}$$

Normal Distribution - Mean and Variance

$$\begin{aligned}M_X(t) &= \frac{e^{\mu t} e^{\frac{t^2 \sigma^2}{2}}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{\omega^2}{2}} d\omega \\&= e^{\mu t} e^{\frac{t^2 \sigma^2}{2}} \\&= \left[1 + \frac{\mu t}{1!} + \frac{\mu^2 t^2}{2!} + \dots \right] \left[1 + \frac{t^2 \sigma^2}{2} + \frac{t^4 \sigma^4}{8} + \dots \right] \\&= \left[1 + \frac{t^2 \sigma^2}{2} + \frac{\mu t}{1!} + \frac{\mu^2 t^2}{2!} + \dots \right]\end{aligned}$$

$$\text{Mean} = \mu'_1 = \text{coeff of } \frac{t}{1!}$$

$$\text{Mean} = \mu$$

$$\mu'_2 = \sigma^2 + \mu^2$$

$$\text{Variance} = \mu'_2 - (\mu'_1)^2$$

$$= \sigma^2 + \mu^2 - \mu^2$$

$$\text{Variance} = \sigma^2$$

Note on Normal Distribution

(i) $B(X: n, p)$ when $n \rightarrow \infty$ and p, q are not small will become a Normal distribution.

(i) $P(X, \lambda)$ when $\lambda \rightarrow \infty$ will become a Normal distribution.

(ii) When $X \sim N(\mu, \sigma^2)$ then $Z = \frac{X - \mu}{\sigma}$ then $Z \sim N(0, 1)$

$$\text{i.e., } f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \quad -\infty \leq z \leq \infty$$

Z is known as a Standard Normal variate with mean 0 and variance 1.

To find probabilities of X we convert X into Z and then make use of standard normal table.

Properties

- (i) The curve is bell shaped and is symmetric about $X = \mu$
- (ii) Mean = Median = Mode = μ
- (iii) Unimodal at $X = \mu$ and Model height
$$= \frac{1}{\sigma\sqrt{2\pi}}$$
- (iv) Skewness $\beta_1 = 0$ and Kurtosis $\beta_2 = 3$
- (v) The points of inflections are at $x = \mu \pm \sigma$

Properties Contd.

- (vi) The X - axis is the asymptote to the curve
- (vii) The mean deviation about mean is 0.8σ
- (viii) Quartile deviation is 0.6745σ
- (ix) If $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$ and X and Y are independent, then $X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$
- (x) Area properties:
 - (a) Total area $P(-\infty < X < \infty) = 1$
 - (b) Area about $X = \mu$: $P(-\infty < X < \infty) = P(\mu < X < \infty) = 0.5$

z table

	Table : Area under standard normal curve from 0 to $\frac{x-\mu}{\sigma}$									
$\frac{x-\mu}{\sigma}$	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.0000	.0040	.0080	.0120	.0159	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1084	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1916	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2258	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2518	.2549
0.7	.2580	.2611	.2642	.2671	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319

z table

1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4430	.4441
1.6	.4452	.4463	.4474	.4485	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4654	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4762	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4865	.4868	.4871	.4875	.5878	.4881	.4884	.4887	.4992
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4980	.4980	.4981
2.9	.4981	.4982	.4983	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990
3.1	.4990	.4991	.4991	.4991	.4992	.4992	.4992	.4992	.4993	.4993

z table

Table entry for z is the area under the standard normal curve to the left of z .

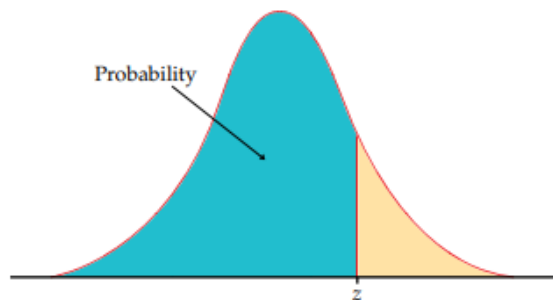


TABLE A

Standard normal probabilities (continued)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319

Problem 1

For a normally distributed variate with mean 1 and standard deviation 3, find the probabilities that

- (i) $3.43 \leq x \leq 6.19$
- (ii) $-1.43 \leq x \leq 6.19$

Problem 1 Contd.

Solution : Given $\mu = 1$ and $\sigma = 3$

(i) When $x = 3.43$, $z = \frac{x - \mu}{\sigma} = \frac{3.43 - 1}{3} = \frac{2.43}{3} = 0.81 = z_1$ (say)

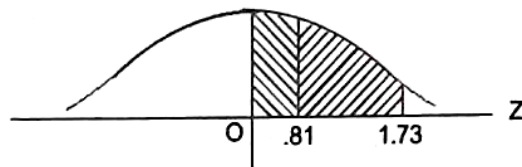
When $x = 6.19$, $z = \frac{x - \mu}{\sigma} = \frac{6.19 - 1}{3} = \frac{5.19}{3} = 1.73 = z_2$ (say)

$$\therefore P(3.43 \leq x \leq 6.19) = P(0.81 \leq z \leq 1.73)$$

$$= |A(z_2) - A(z_1)|$$

$$= |A(1.73) - A(0.81)| = 0.4582^* - 0.2910 \text{ (from tables)}$$

$$= 0.1672 \text{ (Cross hatched area in the figure)}$$



Problem 1 Contd.

$$(ii) \quad \text{When } x = -1.43, z = \frac{x - \mu}{\sigma} = \frac{-1.43 - 1}{3} = -0.81 = z_1 \text{ (say)}$$

$$\text{When } x = 6.19, z = \frac{x - \mu}{\sigma} = \frac{6.19 - 1}{3} = 1.73 = z_2 \text{ (say)}$$

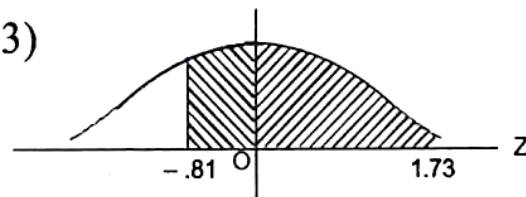
$$\therefore P(-1.43 \leq x \leq 6.19) = P(-0.81 \leq z \leq 1.73)$$

$$= A(z_2) + A(z_1)$$

$$= A(1.73) + A(-0.81)$$

$$= A(1.73) + A(0.81) \quad [\because A(-z) = A(z)]$$

$$= 0.4582 + 0.2910 = 0.7492 = \text{shaded area in the figure.}$$



Problem 2

If X is a normal variate with mean 30 and standard deviation 5, find the probabilities that

- (i) $26 \leq x \leq 40$
- (ii) $X \geq 45$

Problem 2 Contd.

Solution : Given mean, $\mu = 30$ and S.D., $\sigma = 5$

$$(i) \text{ When } x = 26, z = \frac{x - \mu}{\sigma} = \frac{26 - 30}{5} = -0.8 = z_1 \text{ (say)}$$

$$\text{When } x = 40, z = \frac{x - \mu}{\sigma} = \frac{40 - 30}{5} = 2 = z_2 \text{ (say)}$$

$$\begin{aligned} \therefore P(26 \leq X \leq 40) &= P(-0.8 \leq z \leq 2) \\ &= A(z_2) + A(z_1) = A(2) + A(-0.8) \\ &= 0.4772 + 0.2881 = 0.7653 \text{ (from normal distribution tables)} \end{aligned}$$

Problem 2 Contd.

$$(ii) \text{ When } x = 45, z = \frac{x - \mu}{\sigma} = \frac{45 - 30}{5} = 3 = z_1 \text{ (say)}$$

$$\begin{aligned} \therefore P(X \geq 45) &= P(z_1 \geq 3) \\ &= 0.5 - A(z_1) = 0.5 - A(3) \\ &= 0.5 - 0.49865 = 0.00135 \end{aligned}$$

Problem 3

In a normal distribution, 7 % of the items are under 35 and 89 % are under 63. Determine the mean and variance of the distribution.

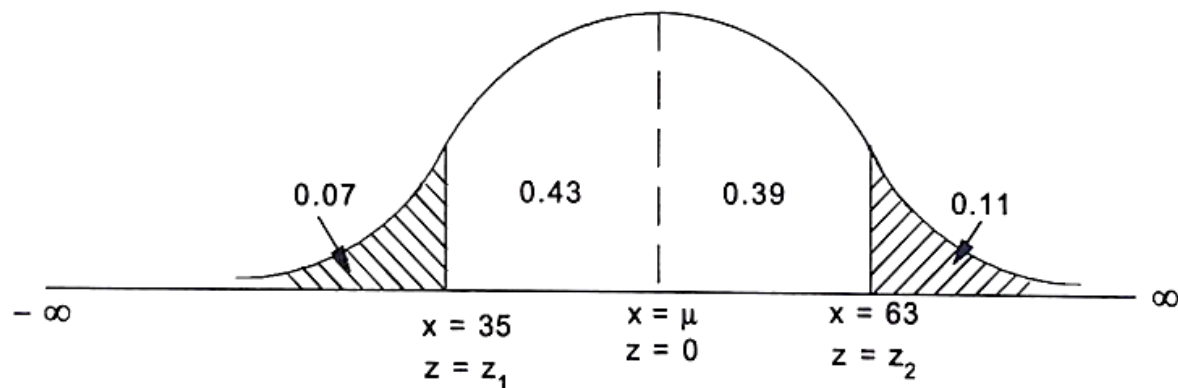
Problem 3 Contd.

Solution : Let μ be the mean (at $z = 0$) and σ the standard deviation of the normal curve. 7% of the items are under 35 means the area to the left of the ordinate $x = 35$.

Given $P(X < 35) = 0.07$ and $P(X < 63) = 0.89$

$$\therefore P(X > 63) = 1 - P(X < 63) = 1 - 0.89 = 0.11$$

The points $X = 35$ and $X = 63$ are shown in the following Figure.



Problem 3 Contd.

$$\text{When } x = 35, z = \frac{x - \mu}{\sigma} = \frac{35 - \mu}{\sigma} = -z_1 \text{ (say)} \quad \dots (1)$$

$$\text{When } x = 63, z = \frac{x - \mu}{\sigma} = \frac{63 - \mu}{\sigma} = z_2 \text{ (say)} \quad \dots (2)$$

From the figure, we have

$$P(0 < z < z_2) = 0.39 \Rightarrow z_2 = 1.23 \quad (\text{from tables})$$

$$\text{and } P(0 < z < z_1) = 0.43 \Rightarrow z_1 = 1.48$$

$$\text{From (1), we have } \frac{35 - \mu}{\sigma} = -1.48 \quad \dots (3)$$

$$\text{From (2), we have } \frac{63 - \mu}{\sigma} = 1.23 \quad \dots (4)$$

Problem 3 Contd.

(4) – (3) gives

$$\frac{28}{\sigma} = 2.71 \Rightarrow \sigma = \frac{28}{2.71} = 10.332$$

From (3), $35 - \mu = -1.48 (\sigma) = (-1.48) (10.332) = -15.3$

$$\therefore \mu = 35 + 15.3 = 50.3$$

$$\text{and variance} = \sigma^2 = 106.75$$

Problem 4

In a normal distribution, 31% of the items are under 45 and 8% are under 64. Determine the mean and variance of the distribution.

Problem 4 Contd.

Solution : Let X be the continuous random variable.

Let μ be the mean and σ the standard deviation.

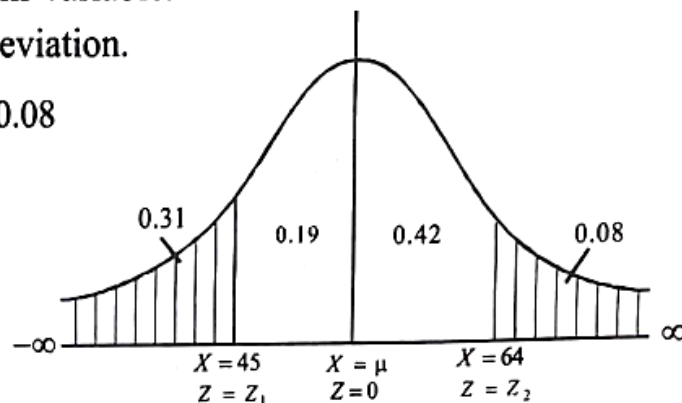
Given $P(X < 45) = 0.31$ and $P(X > 64) = 0.08$

Standard variable, $z = \frac{X - \mu}{\sigma}$

When $X = 45$, let $z = z_1$

so that $z_1 = \frac{45 - \mu}{\sigma}$

... (1)



Problem 4 Contd.

$$\therefore \int_{-\infty}^{z_1} \phi(z) dz = 0.31 \quad \text{or} \quad \int_{-\infty}^0 \phi(z) dz - \int_{z_1}^0 \phi(z) dz = 0.31$$

$$\therefore \int_{z_1}^0 \phi(z) dz = \int_{-\infty}^0 \phi(z) dz - 0.31 = 0.5 - 0.31 = 0.19$$

$$\text{Hence } P(0 < z < z_1) = 0.19 \Rightarrow z_1 = -0.5 \text{ (from table)} \quad \dots (2)$$

$$\text{When } X = 64, \quad z = \frac{64 - \mu}{\sigma} = z_2 \text{ (say)} \quad \dots (3)$$

$$\therefore \int_{z_2}^{\infty} \phi(z) dz = 0.08 \quad \text{or} \quad \int_0^{\infty} \phi(z) dz - \int_0^{z_2} \phi(z) dz = 0.08$$

$$\text{Hence } \int_0^{z_2} \phi(z) dz = \int_0^{\infty} \phi(z) dz - 0.08 = 0.5 - 0.08 = 0.42$$

Problem 4 Contd.

$$\text{Thus } P(0 < z < z_2) = 0.42 \Rightarrow z_2 = 1.4 \text{ (from tables)} \quad \dots (4)$$

$$\text{From (1) and (2), we have } \frac{45 - \mu}{\sigma} = -0.5 \Rightarrow 45 - \mu = -0.5\sigma \quad \dots (5)$$

$$\text{From (3) and (4), we have } \frac{64 - \mu}{\sigma} = 1.4 \Rightarrow 64 - \mu = 1.4\sigma \quad \dots (6)$$

$$(5) - (6) \text{ gives } (45 - \mu) - (64 - \mu) = -0.5\sigma - 1.4\sigma$$

$$\Rightarrow -19 = -1.9\sigma \quad \therefore \sigma = \frac{19}{1.9} = 10$$

$$\text{From (5), } \mu = 45 + 0.5\sigma = 45 + 0.5(10) = 50$$

Hence mean = 50 and standard deviation = 10

Problem 5

Suppose 10 percent of the probability for a normal distribution $N(\mu, \sigma^2)$ is below 35 and 5 percent above 90. What are the values of μ and σ ?

Problem 5 Contd.

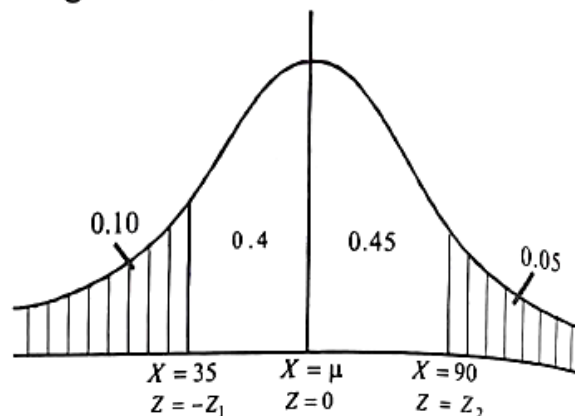
Solution : If $X \sim N(\mu, \sigma^2)$ then we are given $P(X < 35) = 0.10$ and $P(X > 90) = 0.05$. The points $X = 35$ and $X = 90$ are located as shown in Figure.

Standard variable, $z = \frac{X - \mu}{\sigma}$

When $X = 35$, $z = \frac{35 - \mu}{\sigma} = -z_1$ (say)

and when $X = 90$,

$$z = \frac{90 - \mu}{\sigma} = z_2 \text{ (say)}$$



Thus we have $P(0 < z < z_1) = 0.4$ and $P(0 < z < z_2) = 0.45$.

Problem 5 Contd.

Hence from tables, we have $z_1 = 1.3$ and $z_2 = 1.65$

$$\therefore \frac{35 - \mu}{\sigma} = -1.3 \text{ and } \frac{90 - \mu}{\sigma} = 1.65$$

$$\text{Subtracting, we get } \frac{55}{\sigma} = 2.95 \Rightarrow \sigma = \frac{55}{2.95} = 18.64$$

$$\therefore \mu = 35 + 1.3\sigma = 35 + 1.3(18.64) = 35 + 24.24 = 59.24$$

Hence $\mu = 59.24$ and $\sigma = 18.64$

Problem 6

The mean and standard deviation of the marks obtained by 1000 students in an examination are respectively 34.5 and 16.5. Assuming the normality of the distribution, find the approximate number of students expected to obtain marks between 30 and 60.

Problem 6 Contd.

Solution : Given mean, $\mu = 34.5$ and S.D., $\sigma = 16.5$

$$\text{When } x = 30, z = \frac{x - \mu}{\sigma} = \frac{30 - 34.5}{16.5} = -0.27 = z_1 \text{ (say)}$$

$$\text{When } x = 60, z = \frac{x - \mu}{\sigma} = \frac{60 - 34.5}{16.5} = 1.54 = z_2 \text{ (say)}$$

Problem 6 Contd.

$$\begin{aligned}\therefore P(30 \leq x \leq 60) &= P(z_1 \leq z \leq z_2) \\ &= A(z_2) + A(z_1) \quad [\because z_1 < 0 \text{ and } z_2 > 0] \\ &= A(1.54) + A(-0.27) = A(1.54) + A(0.27) \\ &= 0.4382 + 0.1084 \quad [\text{From normal distribution tables}] \\ &= 0.5916\end{aligned}$$

$$\begin{aligned}\therefore \text{The number of students who get marks between 30 and 60} \\ &= 0.5916 \times 1000 = 591.6\end{aligned}$$

Hence 592 students get marks between 30 and 60.

Problem 7

The marks obtained in Mathematics by 1000 students is normally distributed with mean 78% and standard deviation 11%. Determine

- (i) How many students got marks above 90%?
- (ii) What was the highest mark obtained by the lowest 10% of the students?
- (iii) Within what limits did the middle of 90% of the students lie?

Problem 7 Contd.

Solution : Given mean, $\mu = 0.78$ and S.D., $\sigma = 0.11$

(i) When $x = 0.9$, $z = \frac{x - \mu}{\sigma} = \frac{0.9 - 0.78}{0.11} = 1.09 = z_1$ (say)

Hence the number of students with marks more than 90%

$$= 0.1379 \times 1000$$
$$= 137.9 \simeq 138$$

Problem 7 Contd.

- (ii) The 0.1 area to the left of z corresponds to the lowest 10% of the students

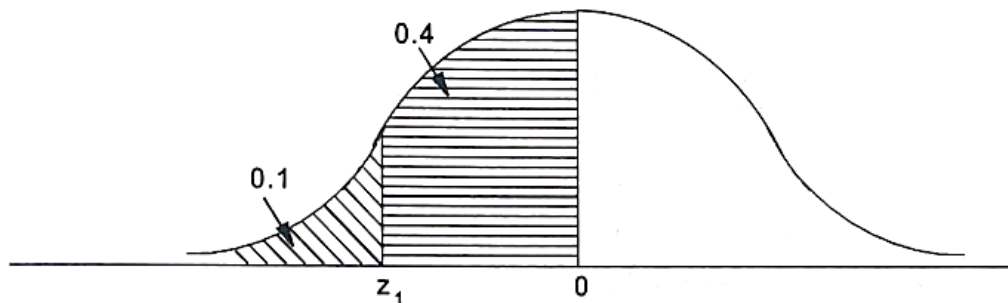


Fig.

From Figure,

$$0.4 = 0.5 - 0.1 = 0.5 - \text{Area from } 0 \text{ to } z_1$$

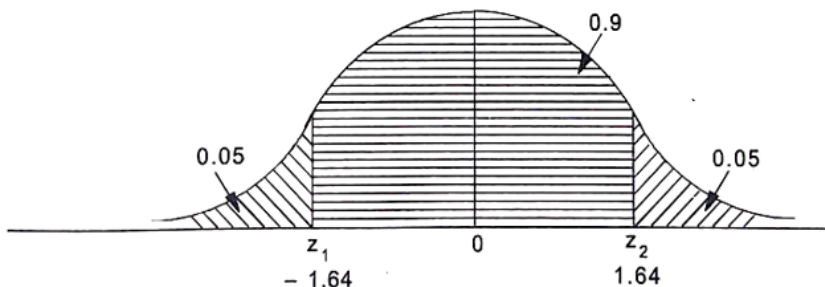
$$\therefore z_1 = -1.28 \text{ (from tables)}$$

$$\text{Thus } -1.28 = \frac{x - \mu}{\sigma} = \frac{x - 0.78}{0.11} \Rightarrow x = 0.78 - 1.28(0.11) = 0.6392$$

Hence the highest mark obtained by the lowest 10% of students
 $= 0.6392 \times 1000 \simeq 64\%$

Problem 7 Contd.

(iii)



Middle 90% correspond to 0.9 area, leaving 0.05 area on both sides. Then the corresponding z 's are ± 1.64 [since if the area from 0 to z is 0.45, then $z = 1.64$]

$$\therefore -1.64 = z_1 = \frac{x_1 - \mu}{\sigma} = \frac{x_1 - 0.78}{0.11}$$

$$\Rightarrow x_1 = 0.78 - 1.64 (0.11) = 0.5996 \text{ or } 59.96\%$$

$$\text{and } 1.64 = z_2 = \frac{x_2 - \mu}{\sigma} = \frac{x_2 - 0.78}{0.11}$$

$$\Rightarrow x_2 = 1.64 (0.11) + 0.78 = 0.9604 \text{ or } 96.04\%$$

Thus the middle 90% have marks in between 60 to 96.

Problem 8

Suppose the weights of 800 male students are normally distributed with mean $\mu = 140$ pounds and standard deviation 10 pounds. Find the number of students whose weights are

- (i) between 138 and 148 pounds
- (ii) more than 152 pounds

Problem 8 Contd.

Solution : Let μ be the mean and σ be the standard deviation. Then $\mu = 140$ pounds and $\sigma = 10$ pounds.

$$(i) \text{ When } x = 138, z = \frac{x - \mu}{\sigma} = \frac{138 - 140}{10} = -0.2 = z_1 \text{ (say)}$$

$$\text{When } x = 148, z = \frac{x - \mu}{\sigma} = \frac{148 - 140}{10} = 0.8 = z_2 \text{ (say)}$$

$$\begin{aligned} \therefore P(138 \leq x \leq 148) &= P(-0.2 \leq z \leq 0.8) \\ &= A(z_2) + A(z_1) = A(0.8) + A(-0.2) \\ &= A(0.8) + A(0.2) \\ &= 0.2881 + 0.0793 = 0.3674 \end{aligned}$$

Hence the number of students whose weights are between 138 pounds and 148 pounds = $0.3674 \times 800 \approx 294$

Problem 8 Contd.

$$(ii) \quad \text{When } x = 152, z = \frac{x - \mu}{\sigma} = \frac{152 - 140}{10} = 1.2 = z_1 \text{ (say)}$$

$$\begin{aligned} \therefore P(x > 152) &= P(z > z_1) \\ &= 0.5 - A(z_1) = 0.5 - A(1.2) \\ &= 0.5 - 0.3849 = 0.1151 \end{aligned}$$

$$\begin{aligned} \therefore \text{Number of students whose weights are more than 152 pounds} \\ &= 800 \times 0.1151 = 92 \end{aligned}$$

Problem 9

The marks obtained in Statistics in a certain examination is found to be normally distributed. If 15% of the students got ≥ 60 marks, 40% got < 30 marks, find the mean and standard deviation.

Problem 9 Contd.

Solution : Let μ be the mean (at $z = 0$) and σ the standard deviation of the normal distribution.

Let the variable X denote the marks in statistics.

Then we are given

$$P(X < 30) = 0.4 \text{ and } P(X \geq 60) = 0.15$$

$$\text{When } X = 30, z = \frac{X - \mu}{\sigma} = \frac{30 - \mu}{\sigma} = -z_1 \text{ (say)} \quad \dots (1)$$

$$\text{When } X = 60, z = \frac{X - \mu}{\sigma} = \frac{60 - \mu}{\sigma} = z_2 \text{ (say)} \quad \dots (2)$$

Problem 9 Contd.

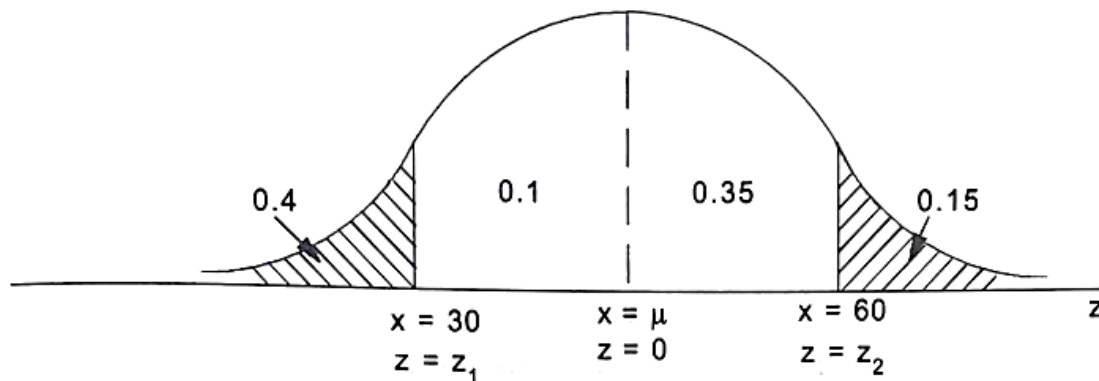


Fig.

$$\therefore P(0 < z < z_2) = 0.5 - 0.15 = 0.35$$

$$\begin{aligned} \text{and } P(0 < z < z_1) &= P(-z_1 < z < 0) \text{ (By symmetry)} \\ &= 0.5 - 0.4 = 0.1 \end{aligned}$$

$\therefore$ From normal tables, we get

$$z_1 = 0.25 \text{ and } z_2 = 1.04$$

Problem 9 Contd.

$$\text{Hence } \frac{30 - \mu}{\sigma} = -0.25 \quad \text{or} \quad \frac{\mu - 30}{\sigma} = 0.25, \text{ using (1)} \quad \dots (3)$$

$$\text{and } \frac{60 - \mu}{\sigma} = 1.04, \text{ using (2)} \quad \dots (4)$$

(3) + (4) gives

$$\frac{30}{\sigma} = 1.29 \quad \therefore \sigma = \frac{30}{1.29} = 23.26$$

$$\begin{aligned} \therefore \mu &= 0.25 \sigma + 30 \quad [\text{From (3)}] \\ &= 0.25 (23.26) + 30 = 35.81 \end{aligned}$$

Hence mean = 35.81 and standard deviation = 23.26

Problem 10

If X is a normal variate, find the area A

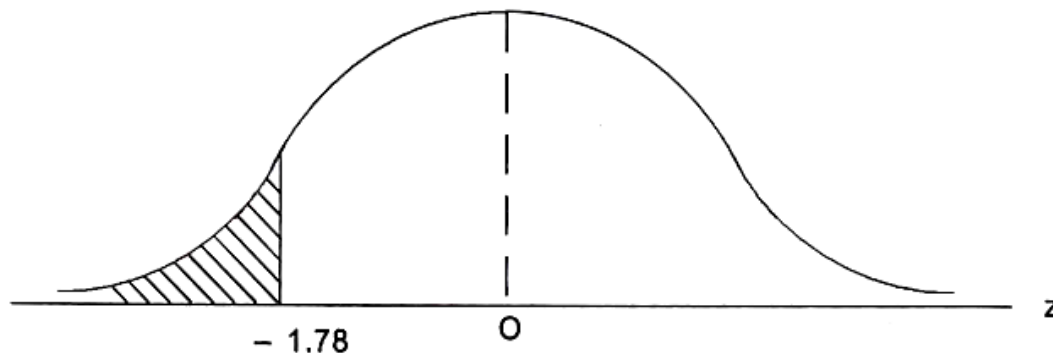
- (i) to the left of $z = -1.78$
- (ii) to the right of $z = -1.45$
- (iii) corresponding to $-0.8 \leq z \leq 1.53$
- (iv) to the left of $z = -2.52$ and to the right of $z = 1.83$

Problem 10 Contd.

Solution :

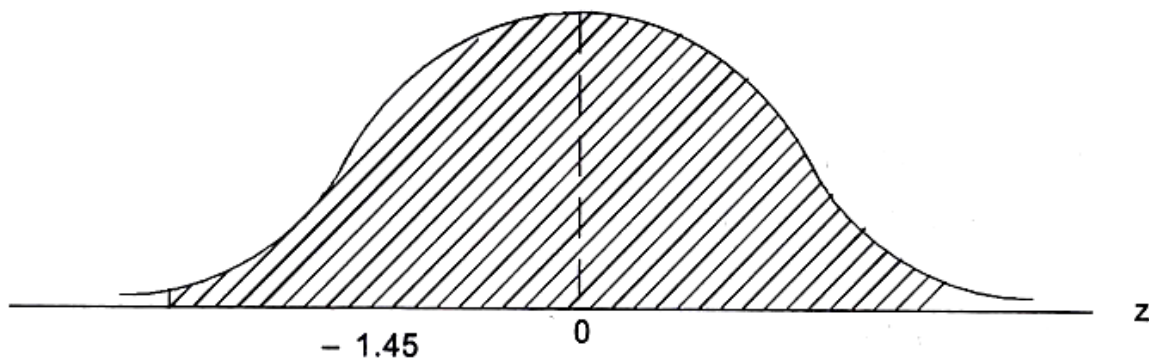
(i) Required area, A

$$\begin{aligned} &= 0.5 - \text{Area (0 to } -1.78) = 0.5 - \text{Area (0 to } 1.78) \text{ (By symmetry)} \\ &= 0.5 - 0.4625 \text{ (from tables)} = 0.0375 \end{aligned}$$



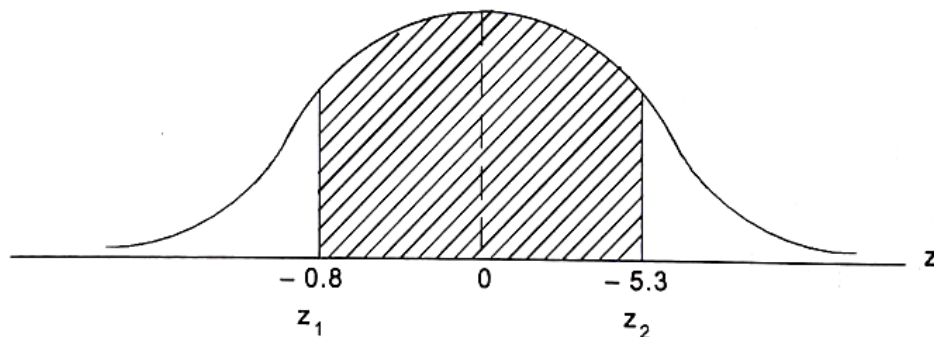
Problem 10 Contd.

(ii) Required area, $A = 0.5 + \text{Area from } 0 \text{ to } -1.45$
 $= 0.5 + \text{Area from } 0 \text{ to } 1.45 \text{ (By symmetry)}$
 $= 0.5 + 0.4265 = 0.9265$



Problem 10 Contd.

$$\begin{aligned} \text{(iii) Required area} &= A(z_2) + A(z_1) \\ &= A(1.53) + A(-0.8) = A(1.53) + A(0.8) \\ &= 0.437 + 0.2881 = 0.7251 \end{aligned}$$

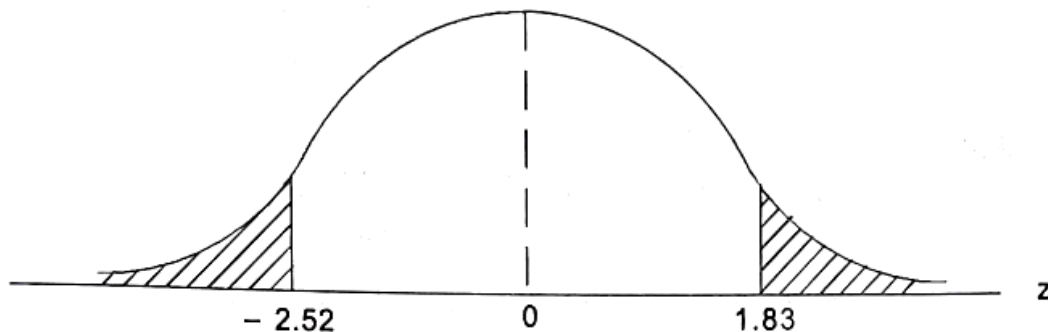


Aliter :

$$\begin{aligned} \text{Required area} &= \text{Area from } (0 \text{ to } -0.8) + \text{Area from } (0 \text{ to } 1.53) \\ &= \text{Area from } (0 \text{ to } 0.8) + \text{Area from } (0 \text{ to } 1.53) \\ &= 0.4370 + 0.2881 = 0.7251 \end{aligned}$$

Problem 10 Contd.

$$\begin{aligned} \text{(iv) Required area} &= [0.5 - \text{Area from 0 to 2.52}] + [0.5 - \text{Area from 0 to 1.83}] \\ &= (0.5 - 0.4941) + (0.5 - 0.4664) = 0.0059 + 0.0336 = 0.0395 \end{aligned}$$



Problem 11

In a sample of 1000 cases, the mean of a certain test is 14 and standard deviation is 2.5. Assuming the distribution to be normal, find

- (i) how many students score between 12 and 15?
- (ii) how many score above 18?
- (iii) how many score below 18?

Problem 11 Contd.

Solution : Let μ be the mean and σ the standard deviation of the normal distribution.
Then we are given

$$\mu = 14 \text{ and } \sigma = 2.5$$

Let the variable X denote the score in a test.

$$(i) \text{ When } X = 12, z = \frac{X - \mu}{\sigma} = \frac{12 - 14}{2.5} = -0.8 = z_1 \text{ (say)}$$

$$\text{When } X = 15, z = \frac{X - \mu}{\sigma} = \frac{15 - 14}{2.5} = 0.4 = z_2 \text{ (say)}$$

Problem 11 Contd.

$$\begin{aligned}\therefore P(12 < X < 15) &= P(-0.8 < z < 0.4) \\ &= A(z_2) + A(z_1) = A(0.4) + A(-0.8) \\ &= A(0.4) + A(0.8) \quad (\text{due to symmetry}) \\ &= 0.1554 + 0.2881 \\ &= 0.4435\end{aligned}$$

$$\begin{aligned}\therefore \text{Number of students score between 12 and 15} \\ &= 1000 \times 0.4435 = 443 \text{ (approximately)}\end{aligned}$$

Problem 11 Contd.

$$(ii) \text{ When } X = 18, z = \frac{X - \mu}{\sigma} = \frac{18 - 14}{2.5} = 1.6$$

$$\begin{aligned} \therefore P(X > 18) &= P(z > 1.6) \\ &= 0.5 - A(1.6) = 0.5 - 0.4452 \\ &= 0.0548 \end{aligned}$$

$$\begin{aligned} \therefore \text{ Number of students score above 18} &= 1000 \times 0.0548 \\ &= 54.8 = 55 \text{ (approximately)} \end{aligned}$$

Problem 11 Contd.

$$(iii) P(X < 18) = P(z < 1.6)$$

$$= 0.5 + A(1.6) = 0.5 + 0.4452 = 0.9452$$

$$\text{Aliter : } P(X < 18) = 1 - P(X > 18) = 1 - 0.0548 = 0.9452$$

$$\therefore \text{ Number of students score below 18} = 1000 \times 0.9452 = 945$$

Problem 12

A sales tax officer has reported that the average sales of the 500 business that he has to deal with during a year is Rs. 36,000 with a standard deviation of 10,000. Assuming that the sales in these business are normally distributed, find

- (i) the number of business as the sales of which are Rs. 40,000
- (ii) the percentage of business the sales of which are likely to range between Rs. 30,000 and Rs. 40,000.

Problem 12 Contd.

Solution : Let μ be the mean and σ the standard deviation of the sales. Then we are given that $\mu = 36000$ and $\sigma = 10000$

Let the variable X denote the sales in the business

$$\text{When } X = 40000, z = \frac{X - \mu}{\sigma} = \frac{40000 - 36000}{10000} = 0.4$$

$$\text{When } X = 30000, z = \frac{X - \mu}{\sigma} = \frac{30000 - 36000}{10000} = -0.6$$

Problem 12 Contd.

$$(i) P(X > 40000) = P(z > 0.4)$$

$$= 0.5 - A(0.4) = 0.5 - 0.1554 = 0.3446$$

Number of business as the sales of which are Rs. 40,000

$$= 500 \times 0.3446 = 172 \text{ (approximately)}$$

$$(ii) P(30000 < X < 40000) = P(-0.6 < z < 0.4)$$

$$= A(0.4) + A(-0.6) = A(0.4) + A(0.6)$$

$$= 0.1554 + 0.2257 = 0.3811$$

$\therefore$ The required percentage of business = 38.11%

Problem 13

If the masses of 300 students are normally distributed with mean 68 kgs and standard deviation 3 kgs, how many students have masses

- (i) greater than 72 kg
- (ii) less than or equal to 64 kg
- (iii) between 65 and 71 kg inclusive.

Problem 13 Contd.

Solution : Let μ be the mean and σ the standard deviation of the distribution. Then

$$\mu = 68 \text{ kgs and } \sigma = 3 \text{ kgs}$$

Let the variable X denote the masses of students

$$(i) \quad \text{When } X = 72, \quad z = \frac{X - \mu}{\sigma} = \frac{72 - 68}{3} = 1.33$$

$$\begin{aligned} \therefore P(X > 72) &= P(z > 1.33) \\ &= 0.5 - A(1.33) = 0.5 - 0.4082 = 0.0918 \end{aligned}$$

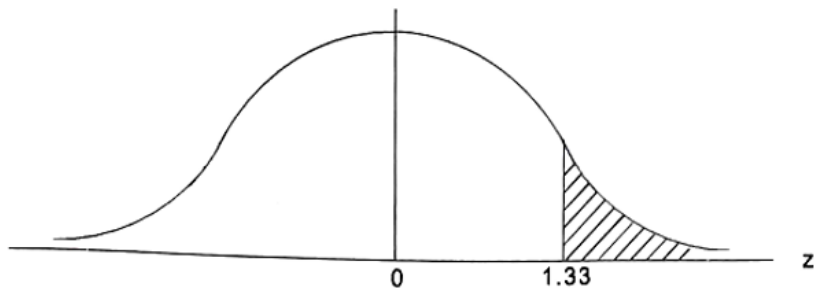


Fig.

Number of students with more than 72 kgs = $300 \times 0.0918 = 28$ (approximately)

Problem 13 Contd.

$$(ii) \text{ When } X = 64, z = \frac{X - \mu}{\sigma} = \frac{64 - 68}{3} = -1.33$$

$$\begin{aligned} \therefore P(X \leq 64) &= P(z \leq -1.33) \\ &= 0.5 - A(1.33) = 0.5 - 0.4082 = 0.0918 \end{aligned}$$

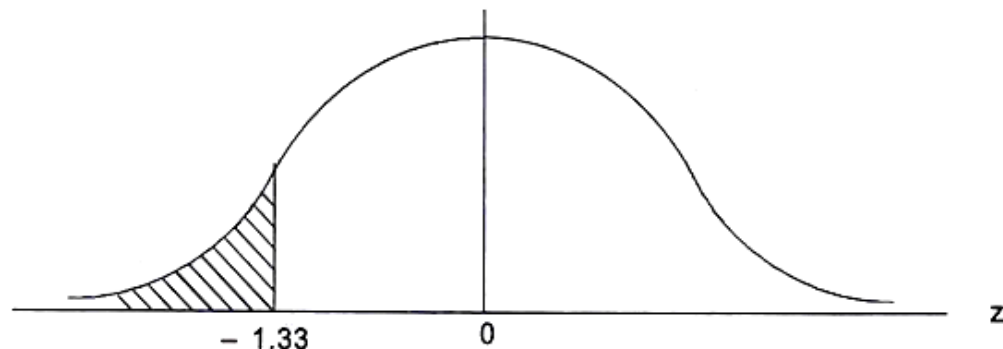


Fig.

Number of students have masses less than or equal to 64 Kg
 $= 300 \times 0.0918 = 28$ (approximately)

Problem 13 Contd.

(iii) When $X = 65$, $z = \frac{X - \mu}{\sigma} = \frac{65 - 68}{3} = -1 = z_1$ (say)

when $x = 71$, $z = \frac{X - \mu}{\sigma} = \frac{71 - 68}{3} = 1 = z_2$ (say)

$$\begin{aligned}\therefore P(65 \leq X \leq 71) &= P(-1 \leq z \leq 1) \\ &= A(z_2) + A(z_1) = A(1) + A(-1) = A(1) + A(1) \\ &= 2.A(1) = 2(0.3413) = 0.6826\end{aligned}$$

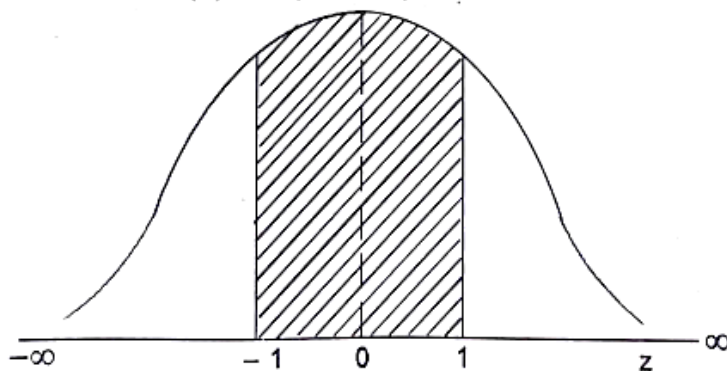


Fig.

$\therefore$ Required number of students $= 300 \times 0.6826 = 205$ (approximately)

Problem 13 Contd.

(iii) When $X = 65$, $z = \frac{X - \mu}{\sigma} = \frac{65 - 68}{3} = -1 = z_1$ (say)

when $x = 71$, $z = \frac{X - \mu}{\sigma} = \frac{71 - 68}{3} = 1 = z_2$ (say)

$$\begin{aligned}\therefore P(65 \leq X \leq 71) &= P(-1 \leq z \leq 1) \\ &= A(z_2) + A(z_1) = A(1) + A(-1) = A(1) + A(1) \\ &= 2.A(1) = 2(0.3413) = 0.6826\end{aligned}$$

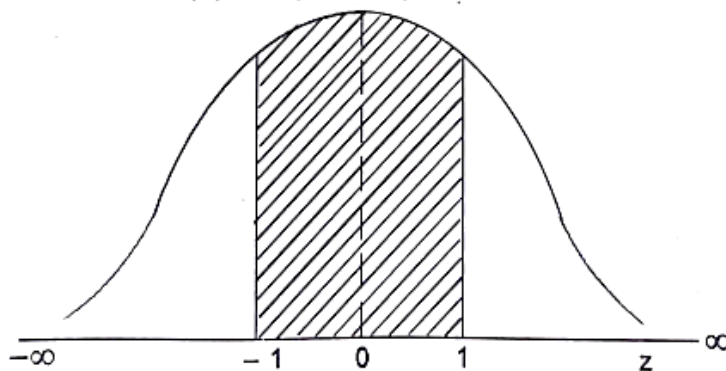


Fig.

$\therefore$ Required number of students $= 300 \times 0.6826 = 205$ (approximately)

Fitting of a Normal Distribution

- (i) Calculate μ and σ of the distribution
- (ii) Find x_i the lower class boundary
- (iii) Find $z_i = \frac{x_i - \mu}{\sigma}$
- (iv) Find Z_i (Z_i) = $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-\frac{z^2}{2}} dz$
- (v) Find $\Delta \phi (Z_i) = \phi (Z_{i+1}) - \phi (Z_i)$
- (vi) Find expected frequency $E_i = N \Delta \phi (Z_i)$

Problem 14

Find expected frequencies for the following data, if its calculated mean and standard deviation are 79.945 and 5.545.

Class	60-65	65-70	70-75	75-80	80-85	85-90	90-95	95-100
Frequency	3	21	150	335	326	135	26	4

Problem 14 Contd.

Given $\mu = 79.945$, $\sigma = 5.545$, and $N = 1000$

Hence the equation of Normal curve fitted to the data is

$$f(x) = \frac{1}{5.545\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-79.945}{5.545}\right)^2}$$

Problem 14 contd.

Theoretical Normal frequencies can be obtained as follows:

Class	Lower Class boundary (X_i)	$z_i = \frac{X_i - \mu}{\sigma}$	$\phi(Z_i) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-\frac{z^2}{2}} dz$	$= \phi(Z_{i+1}) - \phi(Z_i)$	Expected frequencies $N \Delta \phi(Z_i)$
Below 60	$-\infty$	$-\infty$	0	0.001	1
60 - 65	60	-3.59693	0.001	0.0026	$2.6 \approx 3$
65 - 70	65	-2.69522	0.0036	0.0331	$33.1 \approx 33$
70 - 75	70	-1.79351	0.0367	0.15	150
75 - 80	75	-0.89179	0.1867	0.3173	$317.3 \approx 317$
80 - 85	80	0.009919	0.504	0.3146	$314.6 \approx 315$
85 - 90	85	0.911632	0.8186	0.1463	$146.3 \approx 146$
90 - 95	90	1.813345	0.9649	0.0318	$31.8 \approx 32$
95 - 100	95	2.715059	0.9967	0.0032	$3.2 \approx 3$
100 and Over	100	3.616772	0.9999		

Practice Problems I

- ① In an examination, it is laid down that a student passes if he secures 40 percent or more. He is placed in the first, second and third division according as he scores 60% or more marks, between 50% and 60% marks and between 40% and 50% respectively. He gets a distinction in case he secures 75% or more. It is noticed from the results that 10% of the students failed in the examination, whereas 5% of them obtained distinction. Calculate the percentage of students placed in the second division. (Assume normal distribution of marks). (**Ans: 32%**)
- ② A manufacturer knows from experience that the resistance of resistors he produces is normal with mean 100 ohms and standard deviation 2 ohms. What percentage of resistors will have resistance between 98 ohms and 102 ohms? (**Ans: 68%**)

Practice Problems II

- 3 Given that the mean height of students in a class is 158 cms with standard deviation of 20 cms. Find how many students lie between 150 cms and 170 cms, if there are 100 students in the class. (**Ans: 38**)
- 4 The mean and standard deviation of a normal variable are 8 and 4 respectively. Find (i) $p(5 \leq X \leq 10)$ (ii) $p(X \geq 5)$ (**Ans: 0.465, 0.2266**)
- 5 In a test on 2000 electrical bulbs, it was found that the life of a particular make, was normally distributed with an average life of 2040 hours and S.D of 40 hours. Estimate the number of bulbs likely to burn for more than 2140 hours. (**Ans: 12**)

Practice Problems III

- ⑥ If the marks obtained by a number of students in a certain subject are approximately normally distributed with mean 65 and standard deviation 5. If three students are selected at random from this group, what is the probability that at least one of them would have scored above 75? (**Ans: 0.0668**)
- ⑦ If X is normally distributed with mean 2 and variance 0.1, then find $P(|X - 2| \geq 0.01)$? (**Ans: 0.9204**)
- ⑧ The mean height of students in a college is 155 cms and standard deviation is 15. What is the probability that the mean height of 36 students is less than 157 cms. (**Ans: 1.86**)

Practice Problems IV

- 9 The mean inside diameter of a sample of 200 washers produced by a machine is 500 cms with standard deviation 0.005 cms. The purpose of which these washers are intended a maximum tolerance in the diameter 0.495 to 0.505 cms. Otherwise, the washers are considered defective. Determine the percentage of defective washers produced by the machine, assuming the diameters are normally distributed. (**Ans: 31.74%**)
- 10 1000 students have written an examination. The mean of test is 35 and standard deviation is 5. Assuming the distribution to be normal, find
- (i) How many students marks lie between 25 and 40? (**Ans: 89**)
 - (ii) How many students get more than 40? (**Ans: 159**)
 - (iii) How many students get below 20? (**Ans: 1**)
 - (iv) How many students get more than 50? (**Ans: 1**)